

DATA CLEANING & PREPROCESSING

AI & ML Internship – Task 2 Report

1. Introduction

Data cleaning is an important step in data analysis and machine learning. This task focuses on identifying missing values, handling null data, removing unnecessary columns, and preparing the dataset for further analysis using Python libraries such as Pandas, Seaborn, and Matplotlib.

2. Objective

- To identify missing values in the dataset
- To clean null or incomplete data
- To handle numerical and categorical missing values
- To visualize missing data using a heatmap
- To prepare a clean dataset for analysis

3. Dataset Information

Dataset Used: Titanic Dataset
File Format: CSV
Total Rows: 891
Total Columns: 12

4. Technologies Used

Python
Pandas
NumPy
Seaborn
Matplotlib
Jupyter Notebook / VS Code

5. Steps Performed

Step 1: Imported Required Libraries
Step 2: Loaded Dataset using `pd.read_csv()`
Step 3: Checked Missing Values using `df.isnull().sum()`
Step 4: Visualized Missing Values using Heatmap
Step 5: Filled Missing Values using Mean and Mode
Step 6: Removed Cabin Column

Step 7: Saved Cleaned Dataset

6. Output

- Missing values handled successfully
- Dataset cleaned successfully
- Heatmap generated successfully
- Cleaned dataset created

7. Conclusion

This task helped in understanding the importance of data cleaning and preprocessing in machine learning and data analytics. After handling missing values and removing unnecessary data, the dataset became ready for further analysis and model building.